

SQL & Relational Databases

Lesson Guide: Understanding How We Organize & Store Digital Data

1. The Data Explosion: Why Not One Giant Notebook?

Every single day, billions of pieces of data are created across India! From school attendance registers to cricket match scores, library books, and train ticket reservations — there is simply way too much information to write down in a single traditional notebook.

To keep track of everything safely, efficiently, and quickly, we rely on **Smart Digital Storage Systems** called **Databases**.

2. What is a Relational Database?

A **Relational Database** is an organized collection of structured tables that store information neatly.

The Digital Library Analogy

- **Tables = Shelves:** Organized spaces designed to hold specific types of data.
- **Rows = Individual Books:** Each row represents one single record (e.g., one book or one student).
- **Columns = Book Information:** Columns define attributes or details (e.g., Title, Author, Page Count).

3. Tables, Rows, and Columns Explained

Think of a database table just like your school classroom attendance register!

- **Rows (Horizontal):** Represent each individual student's complete record.
- **Columns (Vertical):** Represent specific types of information across all records (e.g., Roll No, Student Name, Date).

- **Cell:** A single box holding one individual piece of data.

Sample Attendance Register Table:

Admission No (Primary Key)	Name	Class	Date
A101	Rahul Sharma	6A	15-Jan
A102	Priya Patel	6B	15-Jan
A103	Rahul Verma	6A	15-Jan
A104	Rahul Singh	6C	15-Jan

4. Why Do We Need a Primary Key?

Imagine your class has three students all named **Rahul**! How do teachers and school computers tell them apart without any mix-ups?

That is why we use a **Primary Key**! A Primary Key is a unique identifier assigned to each row in a table. No two rows can ever have the exact same Primary Key.

 **Real-World Comparison:** Think of a Primary Key like a unique ticket/token at a counter or a unique Admission Number. Even if names repeat, the Primary Key ensures every single record remains distinct and unique!

5. Connecting Tables & Real-World Applications

Relational databases allow different tables to link together using key IDs!

Example: Linking Students to Library Books

A Student Table stores Student IDs and Names. A Library Books Table uses the Student ID to track which student has checked out which book.

Everyday Apps Powered by Databases:

- **IRCTC / Indian Railways:** Uses PNR numbers to manage millions of train bookings.

- **Food & Shopping Apps (Zomato, Amazon):** Tracks orders, customer addresses, and food menu items.
- **Streaming Platforms (Disney+ Hotstar):** Manages user profiles, movie titles, and watch history.

Student Practice Sheet

Topic: SQL & Relational Databases — Review & Exercises



Instructions:

Answer all the questions below based on the SQL & Relational Databases lesson. Show your work clearly where tables are required.

Section A: Short Answer Questions

Question 1

Why can't we use a simple paper notebook to store data for millions of people (like train ticket bookings or school records)? List two reasons.

Question 2

In the "Digital Library" analogy, what do the following database terms correspond to?

- (a) Table = _____
- (b) Row = _____
- (c) Column = _____

Question 3

What is a Primary Key, and why is it essential in a relational database? Give one example of a Primary Key used in schools.

Section B: Applied Table Design Exercise

Question 4: Be a Database Architect!

Imagine you are creating a database table for a **Cricket Team**.

1. Which column would you choose as the **Primary Key** and why?

2. Fill in the blank table structure below with 3 hypothetical players (include attributes like Name, Specialty, and Runs Scored).

_____ (Primary Key)	Player Name	Specialty	Runs Scored

Section C: Real-World Scenario

Question 5

Suppose a food delivery app like Zomato has two separate tables: a Customers Table and an Orders Table. How does the database connect an order to the correct customer? What column should be shared between both tables?